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SUPPLEMENTARY MATERIAL

The Gram Matrix of the Tetrad Equation

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Abstract

The tetrad equation $g = e^T \eta\, e$ is even in the spatial frame, so its spatial realisation M comes with a parity partner $-M$. We study the Gram matrices of the pair. Self-coupling is root-blind and cross-coupling reverses every sign; since the cross-coupling is also unchanged by exchanging the roots, Gram data resolve whether two factors come from the same root or from opposite roots, never which root is which. For the tetrahedral frame, bordering M with a row of ones gives a Hadamard matrix, which yields $M^T M = 4I - J$, the spectral split $I = \frac{1}{4}G + \frac{1}{4}J$ and the rows of M as eigenvectors, at once. The Gram matrix fixes M up to a unique element of $O(3)$, and that element is $-I$ for the two roots: the roots are told apart by orientation alone, which flips because three is odd. The two roots are the parity classes $xyz = \pm 1$ of the cube, and the symmetry group of their compound, the stellated octahedron, is that of one tetrahedron times $\{\pm I\}$. On the connection, exchange of the lower indices splits the 64 components into 40 and 24; the torsion-free condition is the projection onto the 40, and for a metric connection the discarded 24, the torsion, are the only place a connection can tell the roots apart. They carry the root as the frame does: in frame form the torsion reverses with the spatial legs, and in coordinate form its dual, taken with the frame's orientation, reverses sign between the roots, while the torsion-free part has nothing at first order to carry it. Every step is exact. The script checks each one, exhaustively where the cases are finite, and by exact spot checks on rational examples for the converse of Lemma 5, Theorem 3(ii) and Propositions 2 and 3.

Keywords: tetrad, Gram matrix, spatial parity, Hadamard matrix, stellated octahedron, torsion, proof

Setting

A Lorentz tetrad e satisfies the tetrad equation

$$g = e^T \eta\, e, \qquad \eta = \mathrm{diag}(-1, +1, +1, +1). \quad (1)$$

Spatial parity $P = \mathrm{diag}(1, -1, -1, -1)$ acts on the frame index, $e \mapsto Pe$. Since $P^T \eta P = \eta$, the metric is unchanged, $(Pe)^T \eta (Pe) = g$, while $\det(Pe) = -\det e$: the equation has two roots, carried into each other by P, and the metric sees only $(\det e)^2$ [4].

Following [1, 4], the frame is realised in an observer's space by four directions $v_1, \ldots, v_4$, the columns of a matrix $M \in \mathbb{R}^{3\times 4}$, each a fixed linear combination of the spatial legs e^1, e^2, e^3. Every pair of directions bears the same relation and every direction has the same magnitude; these premises force the Gram matrix $M^T M$ and the regular tetrahedron, unique up to $O(3)$ [5]. We take that result as given, scale to $|v_a|^2 = 3$ (every statement below is invariant under a common rescaling), and use the representative

$$M_1 = \begin{pmatrix} +1 & +1 & -1 & -1 \\ +1 & -1 & +1 & -1 \\ +1 & -1 & -1 & +1 \end{pmatrix}. \quad (2)$$

Since each v_a is linear in the spatial legs, P sends every $v_a \mapsto -v_a$, so the second root is realised by $M_2 = -M_1$. Reversing every leg, $e \mapsto -e = PTe$, also reverses the time leg; as in [4], the identification of root with tetrahedron is for future-pointing frames. The legs are labelled throughout: v_a and $-v_a$ belong to the same leg. We write I for the identity, J for the 4×4 all-ones matrix and $\mathbf{1} = (1, 1, 1, 1)^T$.

Part I: The three Gram products

Definition 1. For realisations U, W the Gram product is $G_{UW} = U^T W$, with entries $v_a \cdot w_b$. From the parity pair we form $G_{M_1 M_1}$, $G_{M_2 M_2}$, $G_{M_1 M_2}$ and $G_{M_2 M_1} = G_{M_1 M_2}^T$. We write $G = G_{M_1 M_1}$.

Lemma 1 (Root-blindness)**.** $G_{M_2 M_2} = G_{M_1 M_1}$.

Proof. $(-M_1)^T(-M_1) = M_1^T M_1$. □

Lemma 2 (Cross-coupling)**.** *$G_{M_1 M_2} = G_{M_2 M_1} = -G$. Hence the Gram products take the value G on a same-root pair and $-G$ on a cross-root pair, and each value is unchanged when the two roots are exchanged.*

Proof. $M_1^T(-M_1) = -G = (-M_1)^T M_1$. Exchanging the roots sends $G_{M_1 M_1} \leftrightarrow G_{M_2 M_2}$ and $G_{M_1 M_2} \leftrightarrow G_{M_2 M_1}$, which are equal pairs. □

So the cross-coupling is *root-resolving* in a precise sense: it resolves the *relative* root of two factors. No Gram product says which root a single frame is; Proposition 1 gives the quantity that does.

Lemma 3 (Explicit Gram matrix)**.** *Let* $\hat{H} = \left(\begin{smallmatrix} \mathbf{1}^T \\ M_1 \end{smallmatrix}\right)$. *Then* $\hat{H}\hat{H}^T = 4I$, *hence* $\hat{H}^T\hat{H} = 4I$, $M_1\mathbf{1} = 0$ *and*

$$G = M_1^T M_1 = 4I - J, \qquad M_1 M_1^T = 4I_3, \quad (3)$$

Proof. The rows of M_1 sum to zero and are mutually orthogonal with squared norm 4, by inspection of (2); with the row $\mathbf{1}^T$, of squared norm 4, this is $\hat{H}\hat{H}^T = 4I$. A square matrix with $\hat{H}\hat{H}^T = 4I$ has $\hat{H}^{-1} = \frac{1}{4}\hat{H}^T$, so $\hat{H}^T\hat{H} = 4I$. Its left side is $\mathbf{1}\mathbf{1}^T + M_1^T M_1 = J + G$. □

$\hat{H}$ is the 4×4 Hadamard matrix [6]. Entrywise, G has diagonal 3 and off-diagonal -1; with Lemmas 1 and 2 all three products are determined:

$$G_{M_1M_1} = G_{M_2M_2} = 4I - J, \qquad G_{M_1M_2} = J - 4I. \quad (4)$$

Lemma 4 (Eigenstructure)**.** $\frac{1}{4}G$ *and* $\frac{1}{4}J$ *are complementary orthogonal projections,* $I = \frac{1}{4}G + \frac{1}{4}J$. *Thus* G *has eigenvalue* 0 *on* $\mathbf{1}$ *(closure) and eigenvalue* 4 *with multiplicity* 3 *on* $\mathbf{1}^\perp$, *which is spanned by the rows of* M_1*:* $GM_1^T = 4M_1^T$. *The cross-coupling has eigenvalues* 0 *and* -4 *on the same spaces.*

Proof. $J^2 = 4J$ gives $(\frac{1}{4}J)^2 = \frac{1}{4}J$, and $\frac{1}{4}G = I - \frac{1}{4}J$ is the complementary projection. $GM_1^T = M_1^T(M_1M_1^T) = 4M_1^T$, and the three rows are independent by $M_1M_1^T = 4I_3$. □

Lemma 5 (Invariance and rigidity)**.** *For* $Q \in O(3)$ *the three products are unchanged by* $(M_1, M_2) \mapsto (QM_1, QM_2)$. *Conversely, if* W *has* $W^TW = G$, *then* $W = QM_1$ *for a unique* $Q \in O(3)$, *namely* $Q = \frac{1}{4}WM_1^T$. *For* $W = M_2$, $Q = -I$.

Proof. $(QU)^T(QW) = U^TQ^TQW = U^TW$. The converse is the rigidity lemma of [5], since G has rank 3; multiplying $W = QM_1$ on the right by M_1^T and using $M_1M_1^T = 4I_3$ gives Q. For $W = -M_1$ this is $-I$. □

So, for realisations of rank 3 with labelled legs, the Gram matrix is a complete invariant under $O(3)$, and the two roots are one $O(3)$ orbit: they are related by $-I$, which has determinant -1 and lies outside $SO(3)$.

Proposition 1 (Orientation)**.** *For each triple of legs* $a < b < c$, *the orientation* $\omega_{abc}(M) = \det(v_a, v_b, v_c)$ *is nonzero on* M_1, *invariant under* $SO(3)$, *and satisfies* $\omega_{abc}(M_2) = -\omega_{abc}(M_1)$.

Proof. Any three columns of M_1 are independent, since $\mathbf{1}$ spans the null space of M_1. $\det(Qv_a, Qv_b, Qv_c) = \det Q\, \omega_{abc}$, and $\det(-I_3) = (-1)^3 = -1$. □

This is the spatial counterpart of $\det(Pe) = -\det e$: the roots differ only in orientation, and orientation flips because $D = 3$ is odd.

Part II: Geometric identification

Theorem 1 (Tetrahedra in the cube)**.** *The columns of* M_1 *are the four corners of the cube* $\{\pm 1\}^3$ *with* $xyz = +1$, *and those of* M_2 *the four with* $xyz = -1$. *Each set is a regular tetrahedron centred at the origin, with every edge of squared length* 8. *The two sets are disjoint and together are the eight corners of the cube; their compound is the stellated octahedron* $S = [\, M_1 \mid M_2 \,]$ *[13, 14].*

Proof. Each column of (2) has an even number of entries -1, so $xyz = +1$; negating three coordinates multiplies xyz by $(-1)^3$. The corners with $xyz = +1$ are four in number, so they are exactly the columns of M_1, and the rest are those of M_2. By Lemma 3, $|v_a - v_b|^2 = 3 + 3 - 2(-1) = 8$ for $a \neq b$, and $\sum_a v_a = M_1\mathbf{1} = 0$. □

Following [1, 4], the first tetrahedron is TetA (the identity component under spatial parity) and the second TetB (the parity component).

Theorem 2 (Symmetry and non-isomorphism)**.** *The orthogonal maps permuting the corners of TetA form a group of order 24, acting on the four legs as the whole of* S_4. *Those permuting the corners of* S *form the group*

$$\mathrm{Sym}(S) = \mathrm{Sym}(\mathrm{TetA}) \times \{I, -I\}, \quad (5)$$

of order 48, and every element outside Sym(TetA) *carries TetA onto TetB.*

Proof. A map in Sym(TetA) is fixed by the permutation it induces, as the corners span $\mathbb{R}^3$. Every permutation occurs: by Lemma 5, $M_1\Pi$ has Gram matrix $\Pi^TG\Pi = G$ for any permutation matrix Π, so $M_1\Pi = QM_1$ for some $Q \in O(3)$. Hence the order is 24. Two corners of the cube are at squared distance 8 exactly when they differ in two coordinates, that is, when they have the same xyz. A map $L \in \mathrm{Sym}(S)$ is an isometry, so it carries TetA, four corners at mutual squared distance 8, onto four such corners, all with one value of xyz: onto TetA or onto TetB, and then TetB $= -$TetA onto the other. Those of the first kind form Sym(TetA). If L is of the second kind, $-L$ is of the first, since $-I$ exchanges TetA and TetB; and $-I$ commutes with every map, which gives (5). The verification script confirms the count exhaustively over all 8! relabellings of the corners. □

The extra factor is exactly parity: the compound has twice the symmetry of either

root, and the doubling is the element that exchanges them. Half of the exchanging elements are rotations (for example a quarter turn about a coordinate axis), so TetA and TetB are congruent as unlabelled point sets; as labelled frames they are not, by Proposition 1. This is the caution of [4]: the roots differ in the order of the legs, not in shape.

Part III: The index grid

The affine connection $\Gamma^{\lambda}{}_{\mu\nu}$ has $K^3 = 64$ components. Fixing λ gives a 4×4 *slice* in (μ, ν), with 4 diagonal entries, 6 in the strict upper triangle and 6 in the strict lower triangle. Over the four slices,

$$\underbrace{4 \times 6}_{24} + \underbrace{4 \times 4}_{16} + \underbrace{4 \times 6}_{24} = 64. \tag{6}$$

In the framework's reading of the grid the upper triangle is TetA, the lower triangle TetB and the diagonal shared, and exchanging the triangles is the TetB step [3, 8]; Letter 2 calls the exchange-even sector the identity sector and the exchange-odd sector the parity sector [2].

Lemma 6 (The exchange split)**.** *Let τ exchange the lower indices, $(\tau\Gamma)^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}{}_{\nu\mu}$. Then $\tau^2 = 1$, τ fixes the diagonal and exchanges the two triangles, and $\Pi_+ = \frac{1}{2}(1+\tau)$ is a projection of rank 40 whose kernel, the τ-odd part, has dimension 24 and is the torsion $T^{\lambda}{}_{\mu\nu} = \Gamma^{\lambda}{}_{\mu\nu} - \Gamma^{\lambda}{}_{\nu\mu}$.*

Proof. On a mirror pair with upper entry u and lower entry l, τ swaps u and l; the even combination $\frac{1}{2}(u+l)$ is kept and the odd one $u-l$, the torsion, is removed. Each slice has $4+6$ even and 6 odd directions; times four slices, 40 and 24. □

Written in the coordinates of (6), the torsion-free condition is $l = u$: it keeps the diagonal and one triangle and makes the other a copy, the form in which it reads as keeping shared and TetA and discarding TetB. Independently of which triangle is written down, what is kept is the exchange-even combination $\frac{1}{2}(u+l)$, which does not change when the triangles are exchanged, and what is discarded is the exchange-odd difference $u-l$.

Two cautions. First, the triangles depend on the order in which the coordinates are listed: relabelling them in reverse carries every upper entry into the lower triangle. The exchange τ commutes with every relabelling, and the torsion is a tensor (the inhomogeneous term in the transformation of Γ is symmetric in the lower indices), so the split $40 + 24$ and the condition $T = 0$ are coordinate-independent, while the split $24 + 16 + 24$ is a property of a chosen ordered basis. Second, τ acts on the index slots of the connection and P on the frame; they are different operations, and the only link between them proved here is Theorem 3(iii).

Part IV: The torsion-free projection

Theorem 3 (Torsion-free projection)**.** *(i) The torsion-free condition $T = 0$ is the constraint whose solutions are the image of the projection Π_+: 40 of the 64 components, with the 24 torsion components as the kernel of Π_+. It is not injective, so it is not an isomorphism:*

$$\boxed{16 + 24 = 40 \;\neq\; 64 = 24 + 16 + 24.} \tag{7}$$

(ii) For any metric g, a connection compatible with g is determined by g and its torsion: the differences of two g-compatible connections form a 24-dimensional space, mapped bijectively onto the 24-dimensional space of torsions.

(iii) Hence, if each root is given a g-compatible connection, the two connections differ only through their torsion. A torsion-free metric connection is the Levi-Civita connection of g, which by (1) is the same on both roots.

Proof. (i) is Lemma 6. (ii) All statements are pointwise. The difference Δ of two g-compatible connections satisfies $\Delta_{\lambda\mu\nu} + \Delta_{\nu\mu\lambda} = 0$, with the first index lowered by g: antisymmetry in two of three indices, $4 \times 6 = 24$ dimensions. If its torsion vanishes, Δ is also symmetric in the last two, and a tensor symmetric in one pair of slots and antisymmetric in another vanishes; so $\Delta \mapsto T(\Delta)$ is injective between spaces of equal dimension, and bijective. Its inverse is the contorsion [11]. (iii) The metric is the same on both roots (Setting), so both connections are the Levi-Civita connection of g plus the contorsion of their torsion, which vanishes in the torsion-free case. The verification script checks (ii) exactly for η and for a generic Lorentzian metric. □

Proposition 2 (How a connection follows the root)**.** *Let ω_μ be a Lorentz connection, $\eta\omega_\mu$ antisymmetric, and $\Gamma^{\lambda}{}_{\mu\nu} = (e^{-1})^{\lambda}{}_{a}\big(\partial_\mu e^{a}{}_{\nu} + \omega^{a}{}_{\mu b}\, e^{b}{}_{\nu}\big)$, which is compatible with $g = e^T \eta e$.*

(a) If ω is carried along with the frame, $(e, \omega) \mapsto (Pe, P\omega P)$, then $P\omega P$ is again a Lorentz connection and Γ, with its torsion, is unchanged.

(b) If both roots carry the same ω, the two connections are compatible with the same g and in general differ; by Theorem 3(ii) they differ exactly by a contorsion, with nonzero torsion difference.

Proof. (a) $P^T\eta P = \eta$ keeps $\eta P\omega P$ antisymmetric, and $(Pe)^{-1}(P\partial e + P\omega P\, Pe) = e^{-1}(\partial e + \omega e)$ since $P^2 = 1$. (b) Both are of the stated form with the same g; the script exhibits a rational example with nonzero torsion difference. □

Proposition 3 (How the torsion carries the root)**.** *Let T be the torsion of the connection of Proposition 2(a), and let ϵ be the frame's volume form, $\epsilon_{0123} = \det e$.*

(i) In frame form, $T^a{}_{\mu\nu} = 2\,\partial_{[\mu}e^a{}_{\nu]} + 2\,\omega^a{}_{[\mu|b|}e^b{}_{\nu]}$, the torsion follows the root: $(Pe, P\omega P)$ gives PT^a, its spatial legs reversed as the frame's are. Its coordinate form $(e^{-1})^\lambda{}_a T^a{}_{\mu\nu}$ is the same on both roots.

(ii) $\det(Pe) = -\det e$, so ϵ reverses. The dual $(\star T)^\lambda{}_{\alpha\beta} = \frac{1}{2}\,\epsilon_{\alpha\beta}{}^{\mu\nu}\,T^\lambda{}_{\mu\nu}$ and the axial vector $A^\sigma \propto \epsilon^{\sigma\lambda\mu\nu}T_{\lambda\mu\nu}$ reverse sign between TetA and TetB, and $\star T \neq 0$ whenever $T \neq 0$.

(iii) The totally antisymmetric part of Γ is half that of T, so the torsion-free part has no axial part. More generally a torsion-free connection has no tensor at first order in derivatives (normal coordinates), so at the connection level nothing torsion-free can carry the root.

Proof. (i) $d(Pe) + P\omega P\, Pe = P(de + \omega e)$ since $P^2 = 1$, and $(Pe)^{-1}P = e^{-1}$. (ii) By (i) the coordinate torsion and g are the same on both roots, and ϵ changes sign; the dual is invertible ($\star\star = -1$ on 2-forms in Lorentzian signature). (iii) $\Gamma_{[\lambda\mu\nu]} = \frac{1}{2}T_{[\lambda\mu\nu]}$ because the symmetric part drops out of the total antisymmetrisation; the first derivatives of g vanish at a point in normal coordinates, so no tensor is built from them. The script checks (i)–(iii) on a rational example. □

Remark. Parts I and IV say the same thing at two levels. The metric, and every Gram product, cannot tell the roots apart, as $y = x^2$ cannot tell x from $-x$; neither can any torsion-free metric connection, since it is built from the metric. The frame can: it is the root, and its orientation is the absolute invariant (Proposition 1). At the connection level the only room for the distinction is the 24 torsion components that general relativity sets to zero [9] and Einstein–Cartan theory keeps [10, 11], and Proposition 3 shows that the torsion carries it: in frame form it reverses with the root, and in coordinate form its dual, taken with the frame's orientation, reverses sign. So the torsion is the necessary carrier of the TetA/TetB distinction at the connection level and, whenever it is nonzero, an actual one, as Letter 2 [2, 7] states. Proposition 2(a) says only that the coordinate torsion tensor alone, read without the orientation, is the same on both roots, as y is the same for x and $-x$. Duals of the curvature taken with the orientation also reverse; the statement here is at the connection level, where Letter 2 makes it.

Contributions and prior work

The ingredients are classical: rigidity of Gram matrices under the orthogonal group, the 4×4 Hadamard matrix [12], the stellated octahedron and its symmetry $T_d \times \{\pm I\} = O_h$ [14], the fundamental theorem of Riemannian geometry [9], and the decomposition of a metric connection into Levi-Civita connection and contorsion [11]. What this note adds is their assembly on the two roots of the tetrad equation: the distinction between relative and absolute root, with orientation as the absolute invariant; the compound's symmetry as one root's doubled by the exchange; and the exact form of the statement that torsion is the only place a metric connection can tell the roots apart, and of how it carries them: through the frame's orientation.

Summary

Result	Proves	Requires
Lemma 1	$G_{M_2M_2} = G_{M_1M_1}$	$M_2 = -M_1$
Lemma 2	$G_{M_1M_2} = G_{M_2M_1} = -G$	idem
Lemma 3	$G = 4I - J$	Hadamard
Lemma 4	$I = \frac{1}{4}G + \frac{1}{4}J$	Lemma 3
Lemma 5	Invariance, rigidity	[5]
Prop. 1	Orientation flips	$D = 3$ odd
Thm. 1	$xyz = \pm 1$; cube	Lemma 3
Thm. 2	Symmetry $\times\{\pm I\}$	Lemma 5
Lemma 6	$64 = 40 + 24$ under τ	Indices
Thm. 3	Roots differ only by T	Lemma 6
Prop. 2	Coordinate Γ same on both	$P^2 = 1$
Prop. 3	Torsion carries the root	Prop. 1

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